

When Alignment Erases Disagreement

Preference Collapse and a Four-Valued Alternative for LLMs

Aizierjiang Aiersilan

AAAI 2026 Fall Symposium Series

The George Washington University | alexandera@gwu.edu

Outline

1. Aligning With Whom?
2. Two Results on Preference Collapse
3. Where the Cost Falls
4. A Four-Valued Architecture
5. Governance and Evaluation
6. Conclusion

Aligning With Whom?

Every alignment method answers one question

RLHF fits a scalar reward to pairwise comparisons [10, 26]; **DPO** drops the *explicit* reward model but keeps the same **Bradley–Terry** likelihood [28, 6]; **Constitutional AI** swaps part of the human label stream for AI feedback conditioned on a written constitution [3].

- → A distribution of human judgments → **one scalar** per response.
- → One scalar → **one policy**.

However diverse the population, the pipeline returns a single ordering.

The question

Every deployed method answers **aligning with whom?** — deliberated or not. The answer is encoded in the **estimator**.

Whose values do aligned models express?

The empirical record points in one direction.

- Aligned models track US, some European and South American respondents most closely [15].
- Against the Integrated Values Surveys (**107** countries and territories), **five** successive OpenAI models expressed values resembling English-speaking and Protestant European countries [41].
- Human-feedback tuning shifts expressed views toward the **liberal, high-income and well-educated** [33].
- Alignment widens the gap between American English and Indian or Nigerian English from about **1%** to **17.1%**; one widely used open reward model ranks **99.4%** of other countries more negatively than the US [32].



The effect comes from **alignment itself**, not pretraining alone.

Cultural knowledge: the same pattern

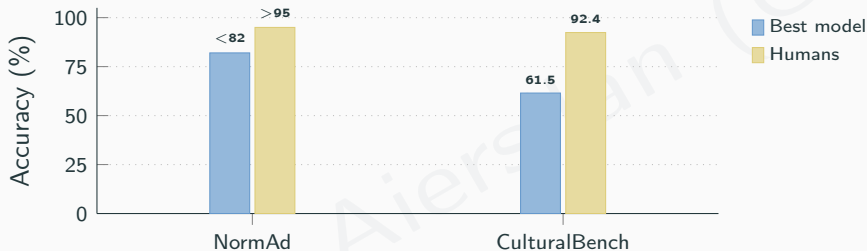


Figure 1: **NormAd**: even when the norms of 75 countries are supplied, the best model stays *below* 82% while humans exceed 95% [29]. **CulturalBench** hard split: frontier accuracy spans 28.7%–61.5% against 92.4% for humans [9].

The standard diagnosis — and what it misses

The usual reading is **representational**: the data lack demographic diversity.

- ✓ Correct, and its remedies are necessary.
- ✗ Incomplete: even **PRISM** — 1,500 participants from 75 countries — skews White, Western, educated, and is English-only [20].

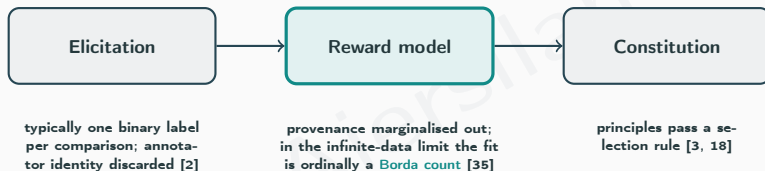
Two failures, kept apart

A **representation** failure: a community is absent from the data.

A **representability** failure: the model class has **no state** that can encode what the community holds.

- 💡 Fix representation alone and you consult the whole world — and still return exactly one answer.

Pluralism is lost at three sites



One reward model cannot represent diverse preferences [8], and it can represent a collective preference only with **no Condorcet cycle** — cycles become almost certain as the response set grows [23].

Two Results on Preference Collapse

The elicitation model

Fix a prompt x and an ordered pair of candidate responses (y_1, y_2) .

- μ is a distribution over **annotator types**; each elicitation draws a type independently.
- Each label records $Z = 1[\text{annotator reports } y_1 \succ y_2]$, and we write $q = \Pr(Z = 1)$.
- Annotator identity is **not retained**, and no annotator labels the same item twice.

The likelihood everything rests on

$$\Pr(y_1 \succ y_2) = \sigma(r(y_1) - r(y_2)), \quad \sigma(\cdot) = \text{logistic}$$

This is the **Bradley–Terry** model that underwrites both **RLHF** and **DPO**.

Proposition 1: two worlds, one dataset

Divided

A fraction q prefer y_1 **with certainty**; the rest prefer y_2 with certainty.

Ambivalent

Every annotator is internally undecided and reports y_1 with probability q .

Indistinguishable

For every n , the labels are i.i.d. Bernoulli(q) under *both* structures, so no test separates them with power exceeding its size.

For $0 < w < n$ the estimate solves $\sigma(\hat{\theta}) = w/n$; hence $\hat{\theta} \rightarrow \text{logit}(q)$ almost surely — and to **zero** when $q = 1/2$.

 *From the labels alone*, half the world calling a response obligatory and the other half forbidden looks exactly like **nobody having a preference**.

The only trace left in the deployed model

Corollary

If the reward attains the per-pair **Bradley–Terry** optimum, then under the KL-regularised **RLHF** objective with coefficient β ,

$$\frac{\pi^*(y_1 | x)}{\pi^*(y_2 | x)} = \frac{\pi_{\text{ref}}(y_1 | x)}{\pi_{\text{ref}}(y_2 | x)} \left(\frac{q}{1 - q} \right)^{1/\beta}$$

The population structure survives only as the scalar q . Under best-of- n reranking, the choice depends only on $\text{sign}(q - \frac{1}{2})$.

The consequence

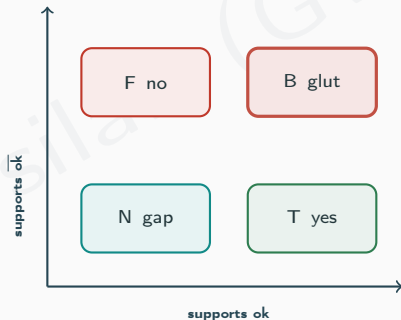
A **99:1** consensus and a **51:49** split induce the **same reranked choice**: the **49%** get an answer their tradition rejects, with no sign that anything was contested. This is the **preference collapse** of [45].

Four states, not two

An open-world mass function over $\Theta = \{\text{ok}, \overline{\text{ok}}\}$ keeps four numbers [39]:

- τ — **support**
- ϕ — **opposition**
- ν — **ignorance** (a **gap**)
- κ — **conflict** (a **glut**)

These are the four values of **Belnap–Dunn** logic [4, 14].



Current pipelines have states for T and F only. Neutrosophic sets keep independent truth, indeterminacy and falsity components [37]; on a binary frame a credal set holds ignorance but has *no state for a glut*, which is why this proposal is paraconsistent [44, 21].

Proposition 2: what a scalar reward throws away

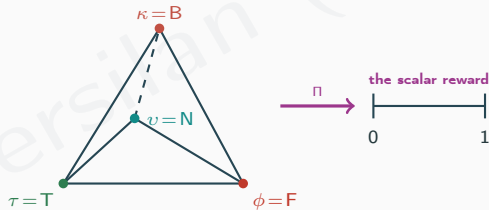
The set of open-world mass functions is the standard **three-simplex**, with the four Belnap values as vertices.

The canonical scalarisation, defined for $\kappa < 1$,

$$\Pi(m) = \frac{\tau + v/2}{1 - \kappa}$$

discards κ by normalisation and splits v evenly.

💡 Every fiber $\Pi^{-1}(p)$, $p \in (0, 1)$, is **two-dimensional**: three degrees of freedom in, one out — **two** are always destroyed.



Normalisation in action

Two communities, nearly categorical and **opposed** [12, 47]:

$$m_1(\{\text{ok}\}) = \mathbf{0.99}, \quad m_1(\Theta) = \mathbf{0.01}; \quad m_2(\{\overline{\text{ok}}\}) = \mathbf{0.99}, \quad m_2(\Theta) = \mathbf{0.01}.$$

Unnormalised combination

- $\emptyset$ (conflict): **0.9801**
- each singleton: **0.0099**
- Θ : **0.0001**

After Dempster's rule ($\div 0.0199$)

- each singleton: **0.4975**
- Θ : **0.005**
- conflict: *gone*

 A near-total contradiction is reported as **near-perfect indifference** — Proposition 1's collapse, by a different operator.

Consensus-seeking tightens the filter

Collective Constitutional AI gathered **1,127** statements and **38,252** votes from a representative sample of **1,002** US adults, keeping only the statements that ranked highest under group-aware consensus — an *effective* threshold of **0.723** [18].

- ✓ Group-aware consensus guards against majoritarianism.
- ✗ A statement one cluster holds central and another rejects scores low *by construction*.

→ The statements marking **irreducible value conflict** are exactly the ones removed.

The same shape appears in deliberation support: participants rated the statements of the AI mediator studied by [42] more informative, clear and unbiased than those of human mediators, and its successful statements incorporated dissent *while respecting the majority position*. A conflict-minimising projection is a virtue for a shared constitutional minimum [30] — and a defect applied to all of value.

Where the Cost Falls

An illustrative case of genuine conflict

Te Hiku Media, an iwi media organisation in Te Hiku o Te Ika, Aotearoa New Zealand, curates a corpus of te reo Māori speech and does **not** release it openly; access is granted case by case under tikanga Māori, and its **Kaitiakitanga License** asserts **guardianship rather than ownership** [19].

Under open-data norms

A recording of a public broadcast is a resource whose openness is a good.

Under kaitiakitanga / CARE

The same recording is a *taonga* — a treasure whose circulation is a matter of **collective authority** [7].

⚠ Suppose the prompt *may this corpus be used to train a commercial speech model* went to a globally stratified pool: the label frequency would be near **one**, because the population holding the restrictive norm is tiny relative to any global pool. The system answers yes, confidently, with no trace of the contestation — and no way for the contesting party to be **decisive over its own data**.

A glut, not a gap

Two features generalise:

- Both parties are **well informed** and internally well supported — so this is B, not N.
- The norms are asymmetric in **population** but not in **normative weight** — which headcount cannot express.

Similar conflicts recur over restricted cultural knowledge, ritual practice, human remains and ancestral land — each **stable under new information**.

Illustrative: the architecture has not been tested on this case. Inclusion in a process is not authority over its outputs [27, 24].

The missing state

No scalar means *permitted under one regime and prohibited under another*.

That state is exactly B.

Value pluralism, not a deficit of facts.

Safety, part one: one threshold cannot serve all

A single acceptability scalar means a **single implicit refusal threshold** — and therefore over-refusal and under-refusal at once, across different populations.

- On **XSTest**, a widely used aligned model fully refuses **38%** of **250** clearly benign prompts under its shipped system prompt, and **14%** without it [31].
- Raising safety data to about **10%** of the instruction mix pushes exaggerated-safety refusals past **50%** of a safe-prompt suite [5].

What four values buy

Not the removal of the trade-off — its **reshaping**: contested and unknown cases route to **distinct behaviours** instead of one side of one threshold.

These figures describe existing scalar-reward systems; they motivate the design, they do not test it.

Safety, part two: an attack surface *without a floor*

Any architecture granting different permissions to different communities invites **false claims of membership** — a **community-claim jailbreak**.

The analogous multilingual attack already works:

- **GPT-4** bypass rate rises from **0.96%** in English to **79.04%** under low-resource translation [46].
- Low-resource inputs are about **3×** as likely to elicit harmful content; deliberate multilingual attacks reach **80.92%** unsafe responses from **ChatGPT** [13].

The design response

A **safety floor** that is closed under community claims — no assertion of identity, tradition, jurisdiction or authorisation *in the prompt* moves an action out of it.

A Four-Valued Architecture

Four coordinates, four regimes

Trigger	Value	System behaviour
τ (support)	T	Answer clearly.
ϕ (opposition)	F	Decline, and state the grounds for refusal.
v (ignorance)	N	Ask clarifying questions, hedge, report calibrated uncertainty.
κ (conflict)	B	Name the disagreement and its parties; return an admissible set of perspectives; defer to the decisive governing body.

Table 1: Current pipelines collapse the last two rows into the first two.

N already has machinery — calibrated abstention signals such as semantic entropy [17]. B is absent today, and it is **not abstention**.

Triggering B — and two honest limits

Two aggregations, neither normalised: the **mixture** $\bar{m} = \sum_c w_c m_c$ describes how the population is distributed; the unnormalised **conjunctive** combination exposes conflict — for evaluations with no internal conflict,

$$\kappa(m, m') = m(\{\text{ok}\}) m'(\{\overline{\text{ok}}\}) + m(\{\overline{\text{ok}}\}) m'(\{\text{ok}\}).$$

B fires when the largest *pairwise* conflict exceeds a published, audited threshold κ^* ; otherwise the dominant coordinate of $\bar{m}$ decides.

Limit 1

κ^* is a **free parameter** still needing a principled derivation.

Limit 2

The pairwise-max rule is **not partition-invariant** — relabelling alone can fire it (*backup slide*).

If each community's evaluation is the weighted average of its members', the mixture *is* invariant under refinement. A partition-invariant conflict statistic is the proposal's main open technical problem.

With acceptability as utility, community evaluations induce a **credal set** over acceptability probability.

- **E-admissibility**: keep a response if it maximises expected utility under *at least one* distribution in the set [21].
- **Maximality** is coarser; **Γ -maximin** selects by lower expectation [44].

The choice among them is a **policy decision about risk** — publish it, don't hide it in a decoding parameter.

Pluriversal admissible set

Responses that *no* recognised community categorically rejects.

- ✓ **Large**: optimise inside it for helpfulness.
- ✓ **Singleton**: give it — while naming the disagreement.
- ✗ **Empty**: **escalate**, don't aggregate.

The safety floor, maintained classically

A core $\mathcal{F}$ of prohibitions sits **outside** the four-valued layer, built as a Rawlsian **overlapping consensus** [30]: baseline members include uplift toward mass-casualty weapons, sexual content involving minors, and targeted incitement to violence.

Three properties

- ✓ Closed under community claims.
- ✓ Attribution via an authenticated governance channel — never inferred from prompt text.
- ✓ The value layer may *restrict* the allowable action set, never *expand* it.

Said plainly

$\mathcal{F}$ is a **residual universalism**. Our claim is not that it is neutral — only that it is **smaller and more auditable** than the status quo.

Worst case under a corrupted community evaluation is **over-refusal**, never a new capability. If a community rejects a prohibition in $\mathcal{F}$, **the prohibition stands**.

What the pipeline must change

No new optimiser — **three** changes.

1. **Elicitation** keeps annotator provenance and lets *both* and *neither* be said separately [2]. By Proposition 1, conflict **cannot be recovered afterwards** from provenance-free labels.
 2. **The reward head** becomes vector-valued or distributional. Dispersion from hidden context is already estimable [35], and policies can be optimised against several reward components [8] — *though dispersion alone does not separate conflict from ignorance*.
 3. **The decision rule** moves from *argmax* to **set selection**.
- 💡 Each component exists. What is missing is a system that keeps **all four coordinates** from elicitation through decoding.

Governance and Evaluation

Sen's paradox as a map of the design space

Sen's theorem, restated for communities

A **policy** maps each profile of community preference orders to an **acyclic** social relation. With at least **two** communities and **three** candidate responses, no policy on the unrestricted domain is both **unanimity-respecting** and **minimally decisive** [34].

Under the idealisations of the Borda characterisation, the scalar pipeline orders responses by **Borda count** [35], which respects unanimity and admits every profile — so it cannot be **minimally decisive**. No minority community is decisive over anything.

Three escapes = three design axes

- Restrict the domain — refuse profiles demanding a breach of $\mathcal{F}$.
- Replace one global policy with context-indexed policies — **subsidiarity**.
- Drop a single social relation and return an **admissible set**.

A known theorem used as a diagnostic analogy, not a result of ours.

Giving governance a technical interface

Three instruments already assert authority over how AI systems handle local values — and none has **a way to attach itself to a model**.

- The **UNESCO Recommendation on the Ethics of Artificial Intelligence**, adopted by **193** member states in November **2021** [43].
 - The African Union's **Continental Artificial Intelligence Strategy**, endorsed in July **2024** [1].
 - Indigenous data governance frameworks, which assert that authority more concretely [7, 19].
- A recognised body can be made **decisive over m_c** for its own community within a specified prompt class: an **auditable, revocable** object in a polycentric arrangement [25].

Three cautions: communities are not monolithic; whoever controls the partition holds real power; an interface usable only by well-capitalised Western institutions reproduces the erasure in a new form [36, 16].

Conflict-preserving evaluation

Existing cross-cultural suites keep regional provenance [15, 29, 9, 20] — but none scores whether an item is **contested**, and the knowledge benchmarks grade against a single correct key.

Two new axes

- **Contestation detection** — scored against provenance-preserving annotation, not a majority key.
- **Gap–glut discrimination** — *people disagree* versus *nobody knows*.

And two more

- **Floor integrity** — is $\mathcal{F}$ invariant under community-claim jailbreaks?
- **Four-way calibration** — do (κ, τ, ϕ, v) match the empirical distribution of judgments?

The last generalises the calibration now applied to v alone.

Conclusion

Objections we take seriously

“This is relativism in better notation”

The floor answers it: classical, non-negotiable, closed under community claims, and only ever *enlarged* by the value layer. Saying *communities differ* when the truth is *prohibited everywhere* is a **misclassification** — which floor-integrity metrics are designed to catch.

“Four numbers per community per item is too expensive”

The ignorance coordinate absorbs unmodelled communities, so the partition can stay **coarse** and be refined incrementally. A developer who cannot afford to know whether its users disagree should not call the system **aligned with them**.

Honest limits: Proposition 1 assumes no annotator labels a pair twice — repeated elicitation breaks the indistinguishability; κ^ is free and the trigger is partition-sensitive; the borrowed numbers are illustrative, and the architecture is not yet implemented.*

What would confirm — or refute — this

Supported if

- ✓ contested and unknown items separate well above a scalar-reward baseline;
- ✓ floor integrity survives adversarial community claims at least as well as a classically aligned model of the same capability;
- ✓ helpfulness survives on uncontested items.

Refuted if

- ✗ the extra coordinates prove unidentifiable in practice;
- ✗ surfacing contestation degrades safety on the floor;
- ✗ the four regimes collapse into two because human evaluators cannot tell gaps from gluts.

💡 Next step: a pilot study of estimability on a small set of contested and unknown items.

Takeaways

1. ★ **It is a representability failure, not only a sampling failure.** Under the **Bradley–Terry** likelihood, deep disagreement and shared ambivalence generate identical data.
2. ★ **The honest evaluation space has four coordinates.** The canonical scalarisation's fibers are **two**-dimensional: conflict is discarded, ignorance is split.
3. ★ **Keep four numbers, return sets, hold a classical floor, and give governance an interface.**

In one sentence

It does not make value conflict disappear — it makes conflict something a deployed system can be honest about.

Part of the **pluralistic alignment** agenda [40, 11]; closest to recent paraconsistent treatments of alignment [38, 22], differing by locating the failure in an **identifiability** result and by keeping the value layer on a classical floor.

African Union.

Continental artificial intelligence strategy: Harnessing AI for Africa's development and prosperity.

African Union Commission, Addis Ababa, 2024.

L. Aroyo and C. Welty.

Truth is a lie: Crowd truth and the seven myths of human annotation.

AI Magazine, 36(1):15–24, 2015.

Y. Bai, S. Kadavath, S. Kundu, A. Askell, J. Kernion, A. Jones, A. Chen, A. Goldie, A. Mirhoseini, C. McKinnon, et al.

Constitutional AI: Harmlessness from AI feedback.

arXiv Preprint arXiv:2212.08073, 2022.

N. D. Belnap.

A useful four-valued logic.

In J. M. Dunn and G. Epstein, editors, *Modern Uses of Multiple-Valued Logic*, pages 5–37. D. Reidel, Dordrecht, 1977.

F. Bianchi, M. Suzgun, G. Attanasio, P. Röttger, D. Jurafsky, T. Hashimoto, and J. Y. Zou.

Safety-tuned LLaMAs: Lessons from improving the safety of large language models that follow instructions.

In *The Twelfth International Conference on Learning Representations*, 2024.

R. A. Bradley and M. E. Terry.

Rank analysis of incomplete block designs: I. the method of paired comparisons.

Biometrika, 39(3/4):324–345, 1952.

S. R. Carroll, I. Garba, O. L. Figueroa-Rodríguez, J. Holbrook, R. Lovett, S. Materechera, M. Parsons, K. Raseroka, D. Rodriguez-Lonebear,

R. Rowe, et al.

The CARE principles for indigenous data governance.

Data Science Journal, 19(1):43, 2020.

- S. Chakraborty, J. Qiu, H. Yuan, A. Koppel, D. Manocha, F. Huang, A. Bedi, and M. Wang.
MaxMin-RLHF: Alignment with diverse human preferences.
In Forty-first International Conference on Machine Learning, 2024.
- Y. Y. Chiu, L. Jiang, B. Y. Lin, C. Y. Park, S. S. Li, S. Ravi, M. Bhatia, M. Antoniak, Y. Tsvetkov, V. Shwartz, et al.
CulturalBench: A robust, diverse and challenging benchmark for measuring LMs' cultural knowledge through human-AI red-teaming.
In Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), pages 25663–25701, 2025.
- P. F. Christiano, J. Leike, T. Brown, M. Martic, S. Legg, and D. Amodei.
Deep reinforcement learning from human preferences.
Advances in Neural Information Processing Systems, 30, 2017.
- V. Conitzer, R. Freedman, J. Heitzig, W. H. Holliday, B. M. Jacobs, N. Lambert, M. Mossé, E. Pacuit, S. Russell, H. Schoelkopf, et al.
Position: Social choice should guide AI alignment in dealing with diverse human feedback.
In Forty-first International Conference on Machine Learning, 2024.
- A. P. Dempster.
Upper and lower probabilities induced by a multivalued mapping.
The Annals of Mathematical Statistics, 38(2):325–339, 1967.
- Y. Deng, W. Zhang, S. J. Pan, and L. Bing.
Multilingual jailbreak challenges in large language models.
In The Twelfth International Conference on Learning Representations, 2024.
- J. M. Dunn.
Intuitive semantics for first-degree entailment and 'coupled trees'.
Philosophical Studies, 29(3):149–168, 1976.

E. Durmus, K. Nguyen, T. Liao, N. Schiefer, A. Askill, A. Bakhtin, C. Chen, Z. Hatfield-Dodds, D. Hernandez, N. Joseph, et al. Towards measuring the representation of subjective global opinions in language models. In *First Conference on Language Modeling*, 2024.

A. Escobar.

Designs for the Pluriverse: Radical Interdependence, Autonomy, and the Making of Worlds, volume 10. Duke University Press, Durham, NC, 2018.

S. Farquhar, J. Kossen, L. Kuhn, and Y. Gal.

Detecting hallucinations in large language models using semantic entropy. *Nature*, 630(8017):625–630, 2024.

S. Huang, D. Siddarth, L. Lovitt, T. I. Liao, E. Durmus, A. Tamkin, and D. Ganguli.

Collective constitutional AI: Aligning a language model with public input.

In *Proceedings of the 2024 ACM Conference on Fairness, Accountability, and Transparency*, pages 1395–1417, 2024.

P.-L. Jones, K. Mahelona, S. Duncan, and G. Leoni.

Kaitiaki: Closing the door on open indigenous data.

International Journal on Digital Libraries, 26(1):1, 2025.

H. R. Kirk, A. Whitefield, P. Röttger, A. Bean, K. Margatina, J. Ciro, R. Mosquera, M. Bartolo, A. Williams, H. He, et al.

The PRISM alignment dataset: What participatory, representative and individualised human feedback reveals about the subjective and multicultural alignment of large language models.

Advances in Neural Information Processing Systems, 37:105236–105344, 2024.

I. Levi.

On indeterminate probabilities.

The Journal of Philosophy, 71(13):391–418, 1974.

- M. Y. Leyva-Vázquez, D. R. Rangel, L. J. Cevallos-Torres, A. M. Pérez, and F. Smarandache.
Aligning with whom? a neutrosophic paraconsistent framework for value alignment under irreducible pluralism.
Preprints.org 202607.2190, 2026.
- K. Liu, Q. Long, Z. Shi, W. J. Su, and J. Xiao.
Statistical impossibility and possibility of aligning LLMs with human preferences: From Condorcet paradox to Nash equilibrium.
The Annals of Statistics, 54(4):2110–2135, 2026.
- S. Mohamed, M.-T. Png, and W. Isaac.
Decolonial AI: Decolonial theory as sociotechnical foresight in artificial intelligence.
Philosophy & Technology, 33(4):659–684, 2020.
- E. Ostrom.
Beyond markets and states: Polycentric governance of complex economic systems.
American Economic Review, 100(3):641–672, 2010.
- L. Ouyang, J. Wu, X. Jiang, D. Almeida, C. Wainwright, P. Mishkin, C. Zhang, S. Agarwal, K. Slama, A. Ray, et al.
Training language models to follow instructions with human feedback.
Advances in Neural Information Processing Systems, 35:27730–27744, 2022.
- M.-T. Png.
At the tensions of south and north: Critical roles of global south stakeholders in AI governance.
In *Proceedings of the 2022 ACM Conference on Fairness, Accountability, and Transparency*, pages 1434–1445, 2022.
- R. Rafailov, A. Sharma, E. Mitchell, C. D. Manning, S. Ermon, and C. Finn.
Direct preference optimization: Your language model is secretly a reward model.
Advances in Neural Information Processing Systems, 36:53728–53741, 2023.

A. S. Rao, A. Yerukola, V. Shah, K. Reinecke, and M. Sap.

NormAd: A framework for measuring the cultural adaptability of large language models.

In *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pages 2373–2403, 2025.

J. Rawls.

Political Liberalism.

Columbia University Press, New York, expanded edition, 2005.

P. Röttger, H. Kirk, B. Vidgen, G. Attanasio, F. Bianchi, and D. Hovy.

XSTest: A test suite for identifying exaggerated safety behaviours in large language models.

In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pages 5377–5400, 2024.

M. J. Ryan, W. Held, and D. Yang.

Unintended impacts of LLM alignment on global representation.

In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 16121–16140, 2024.

S. Santurkar, E. Durmus, F. Ladhak, C. Lee, P. Liang, and T. Hashimoto.

Whose opinions do language models reflect?

In *International Conference on Machine Learning*, pages 29971–30004. PMLR, 2023.

A. Sen.

The impossibility of a Paretian liberal.

Journal of Political Economy, 78(1):152–157, 1970.

A. Siththaranjan, C. Laidlaw, and D. Hadfield-Menell.

Distributional preference learning: Understanding and accounting for hidden context in RLHF.

In *The Twelfth International Conference on Learning Representations*, volume 2024, pages 27448–27471, 2024.

M. Sloane, E. Moss, O. Awomolo, and L. Forlano.

Participation is not a design fix for machine learning.

In *Proceedings of the 2nd ACM Conference on Equity and Access in Algorithms, Mechanisms, and Optimization*, pages 1–6, 2022.

F. Smarandache.

Neutrosophy: Neutrosophic Probability, Set, and Logic. Analytic Synthesis & Synthetic Analysis.

American Research Press, Rehoboth, NM, 1998.

F. Smarandache and M. Y. Leyva-Vázquez.

(T, I, N, F) neutrosophic weak and strong paraconsistency and the epistemics of artificial intelligence: Uncertainty, pluralism, and the decolonial turn in alignment.

Neutrosophic Sets and Systems, 100:221–232, 2026.

P. Smets and R. Kennes.

The transferable belief model.

Artificial Intelligence, 66(2):191–234, 1994.

T. Sorensen, J. Moore, J. Fisher, M. L. Gordon, N. Mireshghallah, C. M. Rytting, A. Ye, L. Jiang, X. Lu, N. Dziri, et al.

Position: A roadmap to pluralistic alignment.

In *Forty-first International Conference on Machine Learning*, 2024.

Y. Tao, O. Viberg, R. S. Baker, and R. F. Kizilcec.

Cultural bias and cultural alignment of large language models.

PNAS Nexus, 3(9):pgae346, 2024.

M. H. Tessler, M. A. Bakker, D. Jarrett, H. Sheahan, M. J. Chadwick, R. Koster, G. Evans, L. Campbell-Gillingham, T. Collins, D. C. Parkes, et al.
AI can help humans find common ground in democratic deliberation.
Science, 386(6719):eadq2852, 2024.

UNESCO.

Recommendation on the Ethics of Artificial Intelligence.

United Nations Educational, Scientific and Cultural Organization, Paris, 2022.

P. Walley.

Statistical Reasoning with Imprecise Probabilities.

Taylor & Francis, 1991.

J. Xiao, Z. Li, X. Xie, E. Getzen, C. Fang, Q. Long, and W. J. Su.

On the algorithmic bias of aligning large language models with RLHF: Preference collapse and matching regularization.

Journal of the American Statistical Association, 120(552):2154–2164, 2025.

Z. X. Yong, C. Menghini, and S. Bach.

Low-resource languages jailbreak GPT-4.

In *NeurIPS 2023 Workshop on Socially Responsible Language Modelling Research*, 2023.

L. A. Zadeh.

On the validity of Dempster's rule of combination of evidence.

Technical Report UCB/ERL M79/24, University of California, Berkeley, 1979.

Thank you!

Questions?

Aizierjiang Aiersilan | The George Washington University | alexandera@gwu.edu

Backup: why the two structures are indistinguishable

In the **divided** structure the randomness in Z comes from the *draw of the annotator*; in the **ambivalent** structure, from the annotator's *internal randomisation*.

Either way $Z \sim \text{Bernoulli}(q)$ with independent draws, so the induced measures on $\{0, 1\}^n$ coincide — and identical laws give identical likelihoods and inferences.

The estimate

With $\theta = r(y_1) - r(y_2)$,

$$\ell(\theta) = w \log \sigma(\theta) + (n - w) \log (1 - \sigma(\theta)),$$

so $\ell'(\theta) = w - n\sigma(\theta)$ and $\ell'' < 0$: for $0 < w < n$ the unique stationary point has $\sigma(\hat{\theta}) = w/n$, and $w/n \rightarrow q$ almost surely.

If $w \in \{0, n\}$ the usual separation case applies and there is no interior maximum.

Backup: the B trigger is partition-sensitive

Let $\kappa^* = 0.5$. Community A is evenly split internally, $m_A(\{\text{ok}\}) = m_A(\{\overline{\text{ok}}\}) = 0.5$; community B has $m_B(\{\overline{\text{ok}}\}) = 0.8$, $m_B(\Theta) = 0.2$.

As two communities

$\kappa(m_A, m_B) = 0.5 \cdot 0.8 = 0.4 < \kappa^* \rightarrow B$
does *not* fire.

Split A in half

$m_{A_1}(\{\text{ok}\}) = 1$, $m_{A_2}(\{\overline{\text{ok}}\}) = 1$ gives
 $\kappa(m_{A_1}, m_B) = 0.8 > \kappa^* \rightarrow B$ *fires*.

- ⚠ Every individual judgment is unchanged; only the labelling moved. When each community's evaluation averages its members', $\bar{m}$ is invariant under refinement — the **pairwise-max conflict statistic** is not.